

Tata Steel Hackathon: Topics & Concepts to Learn

PART 1: ROUND 1 CONCEPTS (Data Science & Machine Learning)

1. Classification Metrics (Precision vs. Recall) [DEEP LEVEL]

- Learn the exact difference. $\text{Recall} = \frac{\text{True Positives}}{(\text{True Positives} + \text{False Negatives})}$. $\text{Precision} = \frac{\text{True Positives}}{(\text{True Positives} + \text{False Positives})}$. Understand why 100% Recall means allowing absolutely 0 False Negatives.

2. Pseudo-Labeling & Semi-Supervised Learning [DEEP LEVEL]

- Learn how to explain this: Using an initially trained model to label unlabeled test data, filtering out only the 'high confidence' predictions, and merging them back into the training data to make the model stronger.

3. Random Forest & Decision Trees (Bagging) [MEDIUM LEVEL]

- Know what a decision tree is. Know that a Random Forest is an 'Ensemble' method that builds hundreds of trees ($n_{\text{estimators}}$) and takes a majority vote. Know that 'max_depth' prevents the tree from growing too deep and memorizing noise (overfitting).

4. Gradient Boosting (XGBoost/LightGBM) [DEEP LEVEL]

- Know the difference between Bagging (Random Forest) and Boosting. Boosting builds trees sequentially, where each new tree tries to correct the mathematical errors of the previous tree.

5. K-Fold Cross-Validation [MEDIUM LEVEL]

- Understand why splitting the training data into 10 parts prevents the model from overfitting or getting 'lucky' on a specific data split.

6. Data Imputation (Median vs Mean) [MEDIUM LEVEL]

- Know why the median is mathematically safer than the mean when dealing with industrial sensor datasets that have extreme outliers.

PART 2: ROUND 2 CONCEPTS (Generative AI & Software Architecture)

1. Retrieval-Augmented Generation (RAG) & Vector Databases [DEEP LEVEL]

- Learn how RAG works step-by-step: 1. Chunking text. 2. Embedding text (converting words to mathematical vectors using models like BAAI). 3. Storing in ChromaDB. 4. Semantic Similarity Search (finding vectors closest to the user's question). 5. Context Injection (giving the text to the LLM).

2. Multimodal Vision Models (Qwen-VL) [DEEP LEVEL]

- Understand that Vision-Language models use a 'Vision Encoder' to break an image down into patches, and a 'Text Decoder' to understand the prompt. Learn how to explain that you forced the model to output structured JSON data.

3. Local LLMs & Inference (Ollama) [MEDIUM LEVEL]

- Learn the concept of 'Quantization'. This is how massive 7-billion parameter models can run locally on edge hardware without needing a giant cloud supercomputer.

4. Asynchronous Backend (FastAPI & SSE) [MEDIUM LEVEL]

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- Know what 'async' and 'await' mean in Python (allowing the server to handle other tasks while waiting for the AI to think). Understand Server-Sent Events (SSE), which is how you stream the AI's response token-by-token to the UI.

5. Database Management (SQLite vs PostgreSQL) [MEDIUM LEVEL]

- Be able to explain that SQLite is a single-file database used for prototyping, while PostgreSQL is a distributed, scalable database used in real enterprise production.

6. Prompt Engineering & Intent Routing [DEEP LEVEL]

- Learn how 'System Prompts' restrict AI behavior. Understand the concept of 'Intent Routing': parsing the user's text to trigger hidden backend functions (like [INVENTORY_REQUEST]) instead of just chatting.